de **mathématique** et d'**informatique**
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Université de Strasbourg

Internship report

The Riemann-Hilbert correspondence

by
Damien Junger

supervised by
Annette Huber-Klawitter

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Je souhaite remercier

« Là-haut, la terre est bleue comme une orange »
Livre ancien, *Mur de la tour d'Ær*

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I Prerequisites

I.1 Category theory

Definition I.1.

A *category* $\mathscr{C}$ is given by the data of :

1. a class $\mathrm{Ob}(\mathscr{C})$, called the *class of objects* of $\mathscr{C}$,
2. for every couple of objects (x, y) of $\mathscr{C}$, a class $\mathrm{Hom}_{\mathscr{C}}(x, y)$, called the *class of morphisms* from x to y,
3. for every triplet of objects (x, y, z) of $\mathscr{C}$, a function

$$\begin{aligned} \mathrm{comp}_{(x,y,z)} : \quad \mathrm{Hom}_{\mathscr{C}}(x, y) \times \mathrm{Hom}_{\mathscr{C}}(y, z) &\longrightarrow \mathrm{Hom}_{\mathscr{C}}(x, z) \\ (f, g) &\longmapsto g \circ f \end{aligned}$$

satisfying the following proprerties :

a. (*Associative law*) for every quartet of objects (w, x, y, z) of $\mathscr{C}$, if $(f, g, h) \in \mathrm{Hom}_{\mathscr{C}}(w, x) \times \mathrm{Hom}_{\mathscr{C}}(x, y) \times \mathrm{Hom}_{\mathscr{C}}(y, z)$, then $(h \circ g) \circ f = h \circ (g \circ f)$,

b. (*Identities*) for every object x of $\mathscr{C}$, there exists an element 1_x of $\mathrm{Hom}_{\mathscr{C}}(x, x)$, so that for every couple of objects (w, y) and every morphisms $(f, g) \in \mathrm{Hom}_{\mathscr{C}}(w, x) \times \mathrm{Hom}_{\mathscr{C}}(x, y)$, then $1_x \circ f = f$ and $g \circ 1_x = g$.

$g \circ f$ is called the *composition* of g with f.

Remark I.1.

1. We will use the following notations : $x \in \mathrm{Ob}(\mathscr{C})$ if x in an object of $\mathscr{C}$, $f : x \to y$ if f is an element of $\mathrm{Hom}_{\mathscr{C}}(x, y)$.
2. If $f : x \to y$, x is called the *domain* or *source* of f and y the *codomain* or *target* of f.
3. For our purpose, it is sufficient to consider that for every $(x, y) \in \mathrm{Ob}(\mathscr{C})^2$, $\mathrm{Hom}_{\mathscr{C}}(x, y)$ is a set. In this case, the category is said to be *locally small*.

Definition I.2.

Let $\mathscr{C}$ be a category.
A category $\mathscr{D}$ is said to be a *subcategory* of $\mathscr{C}$ if the following are satisfied :

1. every object of $\mathscr{D}$ is an object of $\mathscr{C}$,
2. for every $(x, y) \in \mathrm{Ob}(\mathscr{D})^2$, then $\mathrm{Hom}_{\mathscr{D}}(x, y) \subseteq \mathrm{Hom}_{\mathscr{C}}(x, y)$,
3. for every $x \in \mathrm{Ob}(\mathscr{D})$, then $1_x \in \mathrm{Hom}_{\mathscr{D}}(x, x)$,
4. for every $(x, y, z) \in \mathrm{Ob}(\mathscr{D})^3$, every $(f, g) \in \mathrm{Hom}_{\mathscr{D}}(x, y) \times \mathrm{Hom}_{\mathscr{D}}(y, z)$, then $g \circ f \in \mathrm{Hom}_{\mathscr{D}}(x, z)$.

$\mathscr{D}$ is said to be a *full subcategory* of $\mathscr{C}$ if in addition, for every $(x, y) \in \mathrm{Ob}(\mathscr{D})^2$, then $\mathrm{Hom}_{\mathscr{D}}(x, y) = \mathrm{Hom}_{\mathscr{C}}(x, y)$.

I.2 Sheaf theory

In the following, X is a topological space and $\mathcal{T}_X$ will denote its topology category.

Definition I.3.

Let $\mathscr{C}$ be a category (often $\mathscr{C} = \mathscr{S}, \mathscr{A}, \mathscr{R}$...).
A *presheaf* of $\mathscr{C}$ is a contravariant functor $\mathcal{F} : \mathcal{T}_X \to \mathscr{C}$.
Explicitly, it is the data of :

1. for every open set U of X, an object $\mathcal{F}(U)$ of $\mathscr{C}$,
2. for every couple of open sets $V \subseteq U$ of X, a morphism $\mathrm{res}_{U,V} : \mathcal{F}(V) \to \mathcal{F}(U)$ verifying :
    a. for every open set U of X, then $\mathrm{res}_{U,U} = \mathrm{id}_U$,
    b. for every open sets $U \subseteq V \subseteq W$ of X, then $\mathrm{res}_{U,V} \circ \mathrm{res}_{V,W} = \mathrm{res}_{U,W}$.

The objects $\mathcal{F}(U)$ are often called *sections of* $\mathcal{F}$ *over* U.

Remark I.2.

If the context is clear, we will denote $\mathrm{res}_{U,V}(f) = f|_U$ (the notation is justified in the next lines).

Example I.3.

1) The sheaf of real valued continuous functions $C^0(-,\mathbb{R})$ with classical restriction functions $\mathrm{res}(U,V)(f) = f|_U$.
2) If X has an additional structure, the above example can be further specified : if X is a differentiable manifold, we can consider $C^\infty(-,\mathbb{R})$, if X is a complex manifold, $\mathscr{O}(-,\mathbb{C})$ ...
3) If C is an object of $\mathscr{C}$, the constant presheaf $\underline{C}$. We will be intersted in $\underline{\mathbb{Z}}$ or $\underline{\mathbb{C}}$.
4) The sheaf of real valued bounded functions $B(-,\mathbb{R})$.

Presheaves comes from a category point of view, so we need morphisms :

Definition I.4.

A *morphism of presheaves* (over the same space X) $\varphi : \mathcal{F} \to \mathcal{G}$ is a natural transformation.
More explicitly, it is the data, for each $U \in \mathcal{T}_X$, of a morphism $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ of $\mathscr{C}$ such that the following diagram commutes for every $U \subseteq V$:

$$\begin{tikzcd}
\mathcal{F}(V) \arrow[r, "\varphi_V"] \arrow[d, "\mathrm{res}^{\mathcal{F}}_{U,V}"'] & \mathcal{G}(V) \arrow[d, "\mathrm{res}^{\mathcal{G}}_{U,V}"] \\
\mathcal{F}(U) \arrow[r, "\varphi_U"] & \mathcal{G}(U)
\end{tikzcd}$$

Definition I.5.

A *sheaf* $\mathcal{F}$ is a presheaf satisfying the extra conditions for any open cover $\{U_i, i \in I\}$ of an open set U of X :

3. for every $f, g \in \mathcal{F}(U)$ such that : $\forall i \in I,\ f|_{U_i} = g|_{U_i}$, then $f = g$.
4. for every $(f_i) \in \prod_{i\in I} \mathcal{F}(U_i)$ such that : $\forall (i,j) \in I^2,\ f_i|_{U_i \cap U_j} = f_j|_{U_i\cap U_j}$, then there exists

$f \in \mathcal{F}(U)$ with $f|_{U_i} = f_i$ for any $i \in I$.

Remark I.4.

For $U = \emptyset$, it says that $\mathcal{F}(\emptyset)$ is the terminal object of $\mathscr{C}$ (if such one exists). Indeed, we can take the open covering of $\emptyset$ by a family indexed by $\emptyset$ itself, and we then have an arrow $\mathcal{F}(\emptyset) \rightarrow$

Example I.5.

- 1) and 2) are indeed sheaves.
- 3) is not, because of the previous remark.
- 4) is not either, with the conter example $\mathbb{R} = \bigcup_{n\in\mathbb{N}} (-n, n)$ and $f : x \mapsto x$.
- If X is Hausdorff, x_* is a point of X, (S, s) a pointed set, the *skycraper sheaf* is given by $\mathrm{sky}_{x_*}(S) : U \mapsto \begin{cases} S & \text{if } x_* \in U \\ s & \text{otherwise.} \end{cases}$

Definition I.6.

We will denote by PSh_X, respectively Sh_X, the category of presheaves, respectively the full subcategory of sheaves, over X.

One can ask what looks like the behavior of the sheaf very locally, on a point. Since a singleton is in general not a point, we need the following definition.

Definition I.7.

Let $\mathcal{F}$ be a (pre)sheaf.
We define the *stalk* at $x \in X$ as :

$$\mathcal{F}_x := \{(U, s) |\ x \in U,\ U \in \mathcal{T}_X,\ s \in \mathcal{F}(U)\}\big/\sim\ ,$$

where $(U, s) \sim (V, t) \Leftrightarrow (\exists W \subset U \cap V) : s|_W = t|_W$.

Remark I.6.

From a categorical point of view, it is also the direct limit : $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$, where the limit is taken over open set U.

It turns out that this localization principle can be used to transform a presheaf into a sheaf in a natural way (in the sense of what follows).

Definition I.8.

Let $\mathcal{F}$ be a presheaf.
We define for every open set U

$$\mathcal{F}^\sharp(U) = \left\{ s : U \rightarrow \coprod_{x\in U} \mathcal{F}_x \ \middle|\ \begin{cases} \forall x \in U,\ s(x) \in \mathcal{F}_x \\ \exists V \subset U, \exists t \in \mathcal{F}(V) \text{ s.th.} \forall y \in V,\ s(y) = t(y) \end{cases} \right\}.$$

$\mathcal{F}^\sharp$ is called the *sheafification of* $\mathcal{F}$.

Proposition I.1.

$\mathcal{F}^\sharp$ is a sheaf.

Example I.7.

$\mathbb{R}^\sharp$ is the sheaf of locally constant functions, more precisely the sheaf of continuous functions $C^0(-,\mathbb{R})$, where $\mathbb{R}$ is equipped with the discrete topology.

Proposition I.2.

$((-)^\sharp, \mathrm{Forget})$ is an adjoint pair of functors between PSh_X and Sh_X.

Proof

blablabla

$\square$

Example I.8.

Suppose that $\mathscr{C}$ admits kernels and cokernels (this assumption holds whenever kernels and cokernels are written).
For $\varphi : \mathcal{F} \to \mathcal{G}$, we define $\mathrm{Ker}(\varphi)(U) := \mathrm{Ker}(\varphi_U)$, $\mathrm{Im}(\varphi)(U) := \mathrm{Im}(\varphi_U)$ and $\mathrm{Coker}(\varphi)(U) := \mathrm{Coker}(\varphi_U)$.
The first one is always a sheaf, but the two others are not in general. In the following, $\mathrm{Im}(\varphi)$ or $\mathrm{Coker}(\varphi)$ will always denote the sheafification of these presheaves.

Proposition I.3.

A complex $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$ is exact if, and only if the induced complex on stalks $0 \to \mathcal{F}_x \to \mathcal{G}_x \to \mathcal{H}_x \to 0$ is exact for every $x \in X$.

Proof

blablabla

$\square$

Remark I.9.

In general, it is not equivalent of the exactness of the induced sequence over every open set U. One could take as a counterexample the exponential sequence on $\mathbb{C}$:

$$\begin{array}{ccccccccc} 0 & \to & \underline{\mathbb{Z}} & \to & \mathscr{O}_{\mathbb{C}} & \to & \mathscr{O}_{\mathbb{C}}^* & \to & 0, \\ & & & & f & \mapsto & \exp(2\pi i f) & & \end{array}$$

that is not exact over $\mathbb{C}^*$, because the map $\mathscr{O}_{\mathbb{C}^*} \to \mathscr{O}_{\mathbb{C}^*}^*$ is not surjective : the map $z \mapsto z$ does not belong to the image since log is not well defined over $\mathbb{C}^*$.

I.3 Complex geometry

Definition I.9.

Let X be a differentiable manifold of dimension $2n$.
An *holomorphic atlas* is a set $\{(U_i, \varphi_i), i \in I\}$ where $X = \bigcup_{i \in I} U_i$ is an open cover and $\varphi : U_i \to \mathbb{C}^n$ are homeomorphism on their image, such that for all $(i,j) \in I^2$, the *transition map* $\varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \to \varphi_j(U_i \cap U_j)$ is holomorphic.

Such a couple (U_i, φ_i) is called a *local coordinate system*, or just *coordinates*.

We say that two holomorpic atlases $\{(U_i, \varphi_i), i \in I\}$ and $\{(\tilde{U}_j, \tilde{\varphi}_j), j \in J\}$ are *equivalent* if for all $(i,j) \in I \times J, \tilde{\varphi}_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap \tilde{U}_j) \to \tilde{\varphi}_j(U_i \cap \tilde{U}_j)$.

Definition I.10.

A *complex manifold of dimension* n is a real manifold X of dimension $2n$ endowed with an equivalence class of holomorphic atlases.

Remark I.10.

In the following, unless explicitly stated otherwise, the complex manifold will be endowed with an representant of the equivalente class, denoted by $\{(U_i, \varphi_i), i \in I\}$.

Definition I.11.

Let X be a complex manifold and $f : X \to \mathbb{C}$.
We say that f is *holomorphic* if for all local coordinates (U, φ), $f \circ \varphi : \varphi(U) \to \mathbb{C}$ is holomorphic.

Definition I.12.

Let X be a complex manifold.
We denote by $\mathscr{O}_X$ the sheaf of holomorphic function on X, *ie* $\mathscr{O}_X : U \mapsto \{f : U \to \mathbb{C}, f \text{ holomorphic}\}$.

Remark I.11.

$\mathscr{O}_{X,x}$ will denote the stalk of $\mathscr{O}_X$ at x. It coincides with the germs of holomorphic functions at x. *Via* local coordinates (U, φ), with $\varphi(x_*) = 0$ for some $x_* \in U$, we have an isomorphism $\mathscr{O}_{X,x} \simeq \mathscr{O}_{\mathbb{C}^n,0}$. It is an integral domain, and we denote by $Q(\mathscr{O}_{X,x})$ the quotient field associated.

More generally, if the codomain is also a complex manifold, we have the following definition :

Definition I.13.

Let X and Y be complex manifolds and $f : X \to Y$ continuous.
We say that f is *holomorphic* if for all local coordinates (U, φ) on X and (U', φ') on Y, $\varphi' \circ f \circ \varphi : \varphi(U \cap f^{-1}(U')) \to \varphi'(U')$ is holomorphic.

Definition I.14.

Let X be a complex manifold.
A *meromorphic function* on X is a map $f : X \to \bigsqcup_{x \in X} Q(\mathscr{O}_{X,x})$ so that, for all $x_* \in X$, there

exists an open neighborhood $U \subset X$ of x_* and $g, h \in \mathscr{O}_U$ verifying : $(\forall x \in U),\ f_x = \frac{g}{h}$.
The sheaf of meromorphic functions is denoted $\mathcal{K}_X$.

Remark I.12.

If $U \subset X$ is an connected open subset, then $\mathcal{K}_X(U)$ is a field, and if X itself is connected, $\mathcal{K}(X) = \mathcal{K}_X(X)$ is called the function field of X.

Definition I.15.

Let X be a complex manifold and r a positive integer.
A *holomorphic vector bundle* or rank r on X is a complex manifold E equipped with a holomorphic map $\pi : E \to X$ such that :

1. for all $p \in X$, the *fiber* $E_p := \pi^{-1}(p)$ is a complex vector space of dimension r,
2. there exists an open covering $X = \bigcup_{i \in I} U_i$ and biholomorphic maps $\psi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{C}^r$, called the *trivialization maps*, satisfying :
   a. the following diagram commutes :

$$\begin{tikzcd} \pi^{-1}(U_i) \arrow[rr, "\psi_i"] \arrow[dr, "\pi"'] & & U_i \times \mathbb{C}^r \arrow[dl, "\mathrm{pr}_1"] \\ & U_i & \end{tikzcd}$$

   b. The induced map $E_p \to \mathbb{C}^r$ for any $p \in U_i$ is $\mathbb{C}$-linear.

If $r = 1$, we say that (E, π) is a *line bundle*.

Remark I.13.

The travialization maps induce linear isomorphisms $\psi_{i,j} = \psi_i \circ \psi_j^{-1} : U_i \cap U_j \to \mathrm{GL}_r(\mathbb{C})$, and the collection $\{\psi_{i,j}, i, j \in I\}$ is called a cocycle.
In particular, it verifies the conditions :

i. $\psi_{i,i} = \mathrm{id}_{U_i}$,
ii. $\psi_{j,k} \circ \psi_{i,j} = \psi_{i,k}$.

Proposition I.4.

Let X be a complex manifold and let r be a positive integer.
The categories $\mathrm{VectBun}^r(X)$ and $\mathrm{LocFree}_r(X)$ are equivalent.

Proof

We define

$$\mathcal{F} : \begin{array}{cl} \mathrm{VectBun}_r(X) & \longrightarrow \mathrm{LocFree}_r(X) \\ E & \longmapsto \mathcal{F}_E \end{array},$$

where $\mathcal{F}_E$ is the sheaf of sections of E. It is well defined by ??.
It is a functor : given a morphism of vector bundles $f : E_1 \to E_2$, we define for each open set U of X the function

$$\begin{array}{rccl} \varphi_U: & \mathcal{F}_1(U) & \longrightarrow & \mathcal{F}_2(U) \\ & s & \longmapsto & f\circ s \end{array}.$$

φ clearly defines a morphism of sheaves.
In order to construct a quasi-inverse, let $\mathcal{F} \in \mathcal{S}_X$ and $\psi : \mathcal{F}(U_i) \to \mathscr{O}_{U_i}{}^{\oplus r}$ local trivialisations on an open cover $\{U_i, i \in I\}$ of X.

$\square$

Definition I.16.

Let $\mathcal{E}$ be a locally free $\mathscr{O}_X$-modules.
A *connection* is a $\mathbb{C}$-linear application $\nabla : \mathcal{E} \to \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ satisfying the *Liebniz relation* : for any local sections s and local function f, then

$$\nabla(fs) = df \otimes s + f\nabla(s).$$

In particular, it induces, for each integer $k > 0$, a $\mathbb{C}$-linear map

$$\begin{array}{rccl} \nabla^{(k)}: & \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^k_X & \longrightarrow & \mathcal{E} \otimes_{\mathscr{O}_X} \Omega^{k+1}_X \\ & s \otimes \omega & \longmapsto & (-1)^k \nabla s \wedge \omega + s \otimes dw \end{array}$$

A connection ∇ is said to be *flat* (or *integrable*) if the *curvature* $K := \nabla^{(1)} \circ \nabla$ is zero.

Remark I.14.

In dimension 1, we have $\Omega^2_X = 0$, so all connections are flat.

Definition I.17.

We define the category $\mathrm{LocFree}^{\nabla}(X)$ with

— locally free $\mathscr{O}_X$-modules equipped with flat connections as objects,

— morphisms of $\mathscr{O}_X$-modules $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ making

$$\begin{tikzcd} \mathcal{E}_1 \arrow[r, "\nabla_1"] \arrow[d, "\varphi"] & \mathcal{E}_1 \otimes_{\mathscr{O}_X} \Omega^1_X \arrow[d, "\varphi\otimes 1"] \\ \mathcal{E}_2 \arrow[r, "\nabla_2"] & \mathcal{E}_2 \otimes_{\mathscr{O}_X} \Omega^1_X \end{tikzcd}$$

commutes, as morphisms between $(\mathcal{E}_1, \nabla_1)$ and $(\mathcal{E}_2, \nabla_2)$.

II The Riemann-Hilbert correspondence

II.1 A correspondence in algebraic topology

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II.2 Holomorphic version on a complex manifold

Theorem II.1 (Riemann-Hilbert correspondence, complex manifold case).

The fonctor

$$\begin{aligned} F: \quad \mathrm{LocFree}^{\nabla}(X) &\longrightarrow \mathrm{LocSys}(X) \\ (\mathcal{E}, \nabla) &\longmapsto \mathcal{E}^{\nabla} := \mathrm{Ker}(\nabla) \end{aligned}$$

induces an equivalence of categories.
It's inverse is given by

$$\begin{aligned} G: \quad \mathrm{LocSys}(X) &\longrightarrow \mathrm{LocFree}^{\nabla}(X) \\ \mathcal{A} &\longmapsto (\mathscr{O}_X \otimes_{\mathbb{C}} \mathcal{A}, d \otimes 1) \end{aligned}$$

Proof

Step 1 : G is well defined, that is : if $\mathcal{A} \in \mathrm{LocSys}(X)$, then $(\mathcal{E}, \nabla) := (\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X, 1 \otimes d) \in \mathrm{LocFree}^{\nabla}(X)$.

First remark that $d \otimes 1$ is well defined by extension of $\mathscr{O}_X \to \Omega^1_X$ with the tensor product. We also use the identification

$$(\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X) \otimes_{\mathscr{O}_X} \Omega^1_X \simeq \mathcal{A} \otimes_{\mathbb{C}} (\mathscr{O}_X \otimes_{\mathscr{O}_X} \Omega^1_X) \simeq \mathcal{A} \otimes_{\mathbb{C}} \Omega^1_X$$

by super-commutativity of the tensor product.
For a small open set such that $\mathcal{A}|_U \simeq \mathbb{C}^r$, we have $(\mathcal{A} \otimes \mathscr{O}_X)|_U \simeq \mathcal{A}|_U \otimes \mathscr{O}_U \simeq \mathbb{C}^r \otimes \mathscr{O}_U \simeq \mathscr{O}_U^{\oplus r}$.
Finally, we have :

$$\nabla^1 \circ \nabla(a \otimes f) = \nabla^1((a \otimes 1) \otimes df) = \nabla(a \otimes 1) \wedge df + a \otimes d(df) = (a \otimes d1) \wedge df = 0.$$

We again used that the image of $a \otimes f$ in $\mathcal{E} \otimes_{\mathscr{O}_X} \Omega^1_X$ is identified with $(a \otimes 1) \otimes df$.

Step 2 : G is a functor.

If $\phi : A \to B$ is a morphism in $\mathrm{LocSys}(X)$, we have an induced $G(\phi) = \phi \otimes 1 : A \otimes_{\mathbb{C}} \mathscr{O}_X \to B \otimes_{\mathbb{C}} \mathscr{O}_X$ that makes the following commutes :

$$\begin{tikzcd} A \otimes_{\mathbb{C}} \mathscr{O}_X \arrow[r, "\phi\otimes 1"] \arrow[d, "1\otimes d"'] & A \otimes_{\mathbb{C}} \mathscr{O}_X \arrow[d, "1\otimes d"] \\ A \otimes_{\mathbb{C}} \Omega^1_X \arrow[r, "\phi\otimes 1"] & B \otimes_{\mathbb{C}} \Omega^1_X \end{tikzcd}$$

Step 3 : F is well defined, that is : if $(\mathcal{E}, \nabla) \in \mathrm{LocFree}(X)$, then $\mathcal{E}^{\nabla} \in \mathrm{LocSys}(X)$.

This is the most difficult part. Let's sketch out the proof.
The result is equivalent to saying that *locally*, the solutions of our system of differential equations given by ∇ is a complex vector space, with dimension equal to the rank of the system. One can then think of the Cauchy-Lipschitz theorem.

Step 4 : F is a functor.

If $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism in $\mathrm{LocFree}^{\nabla}(X)$, then it induces a morphism of sheaves $\phi : \mathrm{Ker}(\nabla_1) \to \mathcal{E}_2$, and its image is a subset of $\mathrm{Ker}(\nabla_2)$ by the relation $\nabla_2 \circ \phi = (f \otimes 1) \circ \nabla_1$.

Step 5 : $GF \simeq 1_{\mathrm{LocFree}^{\nabla}(X)}$.

Let $(\mathcal{E}, \nabla) \in \mathrm{LocFree}^{\nabla}(X)$. Set

$$\alpha_{(\mathcal{E},\nabla)} : \begin{array}{ccl} \mathrm{Ker}(\nabla) \otimes_{\mathbb{C}} \mathscr{O}_X & \longrightarrow & \mathcal{E} \\ s \otimes f & \longmapsto & fs \end{array} .$$

It is clearly well defined and for $s \otimes f \in \mathrm{Ker}(\nabla) \otimes_{\mathbb{C}} \mathscr{O}_X$, we have :

$$\nabla(\alpha(s \otimes f)) = \nabla(fs) = s \otimes df = \alpha(s \otimes 1) \otimes df = (\alpha \otimes 1)((1 \otimes d)(s \otimes f)),$$

so φ define a morphism in the category $\mathrm{LocFree}^{\nabla}(X)$.
Better, it defines a natural transformation. Let $\phi : (\mathcal{E}_1, \nabla_1) \to (\mathcal{E}_2, \nabla_2)$ a morphism in $\mathrm{LocFree}^{\nabla}(X)$. Then,

$$\phi(\alpha_{(\mathcal{E}_1,\nabla_1)}(f \otimes s)) = \phi(fs)$$

and

$$\alpha_{(\mathcal{E}_2,\nabla_2)}(GF(\phi)) = \alpha_{(\mathcal{E}_2,\nabla_2)}(F(\phi)(s) \otimes f) = f\phi(s).$$

This is equal because ϕ is a morphism of $\mathscr{O}_X$-modules.
We define an inverse $\beta_{(\mathcal{E},\nabla)}$ locally. For an open set U such that $\mathcal{E}_U \simeq \mathscr{O}_U^{\oplus r}$, we set $(s_1, \ldots, s_r)$ a basis of $\mathcal{E}(U)$. Then a section $\sigma \in \mathcal{E}(U)$ is uniquely written as $\sigma = \sum_{i=1}^r f_i s_i$, and we define $\beta_{(\mathcal{E},\nabla)}(\sigma) = \sum_{i=1}^r f_i \otimes s_i$. It clearly does not depend on the basis, since two basis differ from one another by a matrix with constant coefficients, and therefore commutes with the tensor product.
Still localy, we have $\alpha_{(\mathcal{E},\nabla)}\ circ\beta_{(\mathcal{E},\nabla)} =$

Step 6 : $FG \simeq 1_{\mathrm{LocSys}(X)}$.

We clearly have for a local system $\mathcal{A}$ on X, $(\mathcal{A} \otimes_{\mathbb{C}} \mathscr{O}_X)^{1 \otimes d} \simeq \mathcal{A} \otimes_{\mathbb{C}} \mathbb{C} \simeq \mathcal{A}$, and it behaves well with morphisms, such that it defines an isomorphism of functors.

□

II.3 Meromorphic version on a complex manifold

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blabla [Huy04]

Références

[Huy04] D. Huybrechts. *Complex geometry, an introduction*. Springer, 2004.